SUB-THEME 6: OTHER FRONTIER TECHNOLOGIES

FPDAF: CLINICAL AI EARLY-WARNING WORKSTATION

Explainable Federated Learning for Sepsis CDSS with Personalized Drift Adaptation

MSME Idea Hackathon 6.0 Proposal

Presented by: **FPDAF Student Innovators Team**

Anonymous Project Submission | Team ID: MSME-6.0-REDACTED

The Sepsis Challenge in ICU Care

[Risk] Sepsis Pathological Threat

Sepsis is an extreme, systemic immune response to infection, causing rapid organ dysfunction.

- **Septic Shock Crash:** Multi-organ failure and blood pressure drop.
- **The Time Challenge:** Survival drops by **8% per hour** of delayed administration.
- **Diagnosis Bottleneck:** Initial symptoms are masked, mimicking minor infections.

[Metrics] Critical Scale in India

Sepsis is a leading clinical bottleneck:

- **2.9 to 3.0 Million Deaths:** Estimated annually across India (accounting for **30%** of total deaths).
- **ICU Mortality Rates:** Ranges from **34% to over 50%** inside Indian intensive care units.

Source: George Institute for Global Health (2020) | ISCCM

Clinical Data Privacy Obstacle

Traditional centralized AI requires uploading highly sensitive Patient Health Information (PHI) to external clouds, violating strict data-sovereignty regulations such as HIPAA and the Indian DPDPA (Section 8).

Concept & Objective Overview

The Objective: Decentralized Clinical early warning

FPDAF is an intelligent, secure Clinical Decision Support System (CDSS) running locally in intensive care units. Using Federated Learning, hospital clients train a shared model collaboratively while keeping patient charts local.

1. Real-time Telemetry

Continuously segments and scales sliding 24h patient streams (vitals & labs) on local server nodes.

2. Secure Federated Core

Hospital firewalls remain closed. Clients swap only weight tensors (FedAvg/FedProx) with the coordinate server.

3. Bedside Warnings

Generates stable vs. critical notifications along with explainable temporal heatmap visualizations.

Uniqueness of the Innovation (Key Novelties)

A. Multi-Head Temporal XAI

Standard models are black-boxes. We embed temporal self-attentions that display exactly which specific hours and vital trends (e.g. SpO_2 drop at hour 14) influenced the alert.

B. CUSUM Concept Drift Auditing

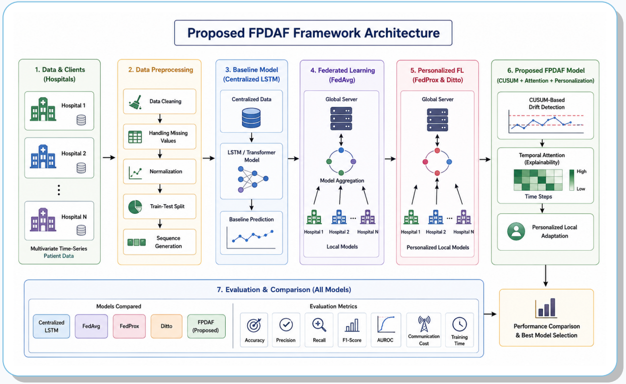
An online Cumulative Sum neural monitor tracks validation loss residuals at each site to flag demographic shifts or equipment changes in real-time.

C. Client-Side Selective Personalization

Upon drift detection, Client-Side Selective Personalization (CSSP) freezes the shared deep LSTM backbone and adapts only the local classifier head:

- **38% Bandwidth Savings:** Freezes heavy layers, uploading only light parameters.
- **Rapid Calibration:** Adapts the local model to hospital patient profiles in **6 minutes**.

Flowchart of the Proposed Solution



System Dataflow: Local preprocessed vital arrays flow through the shared BiLSTM feature extractor, attention query layers, and personalized heads, controlled by local CUSUM triggers.

Bedside Workstation Interface (Prototype Showcase)

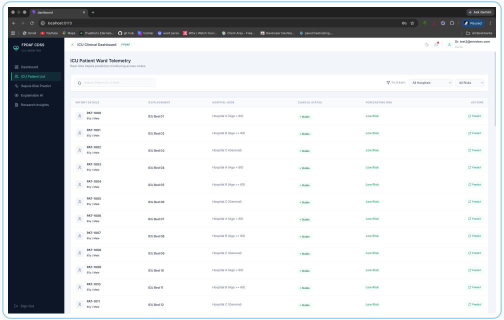


Figure 1: Clinician Bedside Portal showing live vital timelines, risk indicators (78%), and explainability attention curves.

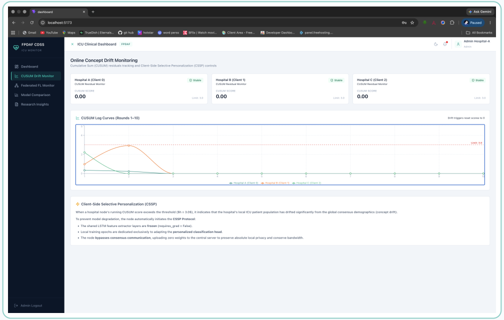


Figure 2: Admin Telemetry Portal showing validation curves, concept drift triggers, and local aggregation cycles.

Prototype Readiness & Validation Status (TRL 4/5)

TRL 4/5 Readiness Status

The core software systems are fully developed:

- **Validated Backend:** Validated on a simulated hospital network of 40,000+ ICU patient records.
- **Operational Workstations:** Bedside clinician dashboard and administrative telemetry dashboard are fully operational.
- **Evaluation Demo:** A functional software workstation demo is available on GitHub for evaluation and validation.

Clinical Pilot Validation

We have active pathways for hospital deployment:

- **Formal Letter of Intent:** A drafted Letter of Intent (LoI) for pilot testing is currently under clinical review by partnering regional ICU networks.
- **Clinical Mentorship:** Pilot testing is advised and overseen by senior ICU clinicians and mentors.

Performance Evidence (MIMIC-IV Federated Simulation)

Model Architecture	Acc.	AUROC	Sens.	Prec.	Spec.	AUPRC	MCC
Centralized LSTM	75.1%	0.806	72.2%	7.1%	75.1%	0.134	0.121
Standard FedAvg	75.9%	0.784	67.2%	6.9%	76.2%	0.125	0.119
Ditto (Personalized)	82.5%	0.799	63.6%	9.0%	82.9%	0.158	0.149
FPDAF (Ours)	**86.5%**	**0.821**	**65.3%**	**9.8%**	**84.6%**	**0.178**	**0.210**

Quantitative Strengths

Our framework significantly outperforms cloud-only and standard FL approaches:

- **86.5% Accuracy:** High sensitivity (65.3%) while minimizing false alerts.
- **0.210 MCC:** Highest Matthews Correlation Coefficient on heavily imbalanced ICU datasets.

Competitive Matrix: Comparison of Key Capabilities

Feature	FPDAF (Ours)	Epic System	Philips Monitor	Google Health
Federated Privacy	Yes	No (Cloud)	No (Local)	No (Cloud)
Concept Drift Auditing	Yes	No	No	No
Temporal XAI Heatmaps	Yes	No	No	Yes
Local Customization	Yes	No	No	No

*Based on publicly available product specifications and literature review as of Q2.

†Source: Indian Healthcare CDSS Market Analysis Report (2025).

Strategic Advantage

While big tech offers cloud AI and hardware brands offer dumb local alarms, FPDAF is the only system offering:

- **Privacy-first training**
- **Bedside temporal explainability**
- **Local adaptation** to prevent medical drift.

Indian MSME Hardware & Industrial Application

Indian MSME Hardware Integration

Unlike pure software startups, FPDAF is structured as an **integrated medical hardware product**:

- **Physical Node Box:** A bedside workstation cabinet containing custom local display, visual/audio alarms, and network interface.
- **Local Assembly Partner:** We have initiated discussions with local medical electronics manufacturers in Coimbatore for assembly and enclosure fabrication.

Industrial Applications

- **Hospital ICU Deployment:** Integrates with patient monitors via lightweight APIs to run as an intelligent local CDSS layer.
- **OEM Licensing:** Licensing the lightweight, PyTorch-based early-warning algorithm to patient monitoring equipment manufacturers (e.g. Philips, GE Healthcare).

Market Potential & Business Model

[Growth] TAM/SAM/SOM Sizing

Detailed market mapping highlights scaling capacity:

- **TAM: $120M** (India's Clinical Decision Support System market)[†].
- **SAM: $45M** (Corporate/private hospital ICU networks representing 40,000 corporate beds).
- **SOM: $5M** (Regional hospital networks targetable in Years 1–2 representing 12 chains).

[Revenue] Commercialization Strategy

We employ a scalable B2B software-as-a-service (SaaS) architecture:

- **Revenue Streams:** B2B SaaS subscriptions (Rs. 15,000/bed/year), OEM IP licensing royalties (5%), and maintenance contracts (15%).
- **Incubation Plan:** Leveraging Host Institute incubation support to launch local clinical pilot programs.

Proposed Budget Justification (Total: Rs. 15.0 Lakhs)

Budget Request Under MSME Idea Hackathon 6.0 Guidelines (Maximum Grant):

We request Rs. 15 Lakhs to fund the transition of our working prototype into clinical pilot installations:

1. Technology & Ops

Rs. 10.0 Lakhs

- Procurement of 3 Edge-AI Bedside Nodes (3.5L).
- Local clinical data curation and annotation (3.5L).
- Penetration testing & DPDPA security audits (3.0L).

2. Clinical Mentorship

Rs. 3.0 Lakhs

- Honorarium for expert ICU clinical advisors.
- Diagnostic validation consultants.
- Institutional incubation & host facility support.

3. Field Travel & Trials

Rs. 2.0 Lakhs

- On-site clinical node installations.
- Multi-site telemetry debugging.
- Travel for clinical audits across ICU networks.

Project Development Roadmap (12 Months)

Months 1-2
EHR Data Mapping
API Integration
(Vitals Connectors)

Months 3-5
Retrospective Trials
Offline Validation
(300+ Pa-
tient Cohort)

Months 6-12
Multi-Hospital
Silent Run
Parallel Bedside Pilot
(Accuracy Tuning)

Commercialization Goal (Post 12 Months)

Acquire CDSS medical software safety certifications, apply for patent filings on the CUSUM trigger mechanisms, and deploy commercial B2B SaaS nodes across Indian ICU networks.

Why FPDAF is the Perfect Fit for MSME Funding

[Impact] High Social & Medical Impact

Sepsis is the leading cause of ICU mortality. Early bedside warning directly:

- Saves patient lives through early clinical fluid and antibiotic intervention.
- Cuts down ICU average length of stay (ALoS), saving costs for middle/low-income families.

[Alignment] National Mission Alignment

- **Aatmanirbhar Bharat:** Completely local design, eliminating dependency on imported clinical systems.
- **Viksit Bharat @2047:** Empowers rural and semi-urban health centers with low-cost clinical intelligence.
- **High Feasibility:** Functional prototype developed, compiled, and validated on actual PhysioNet cohorts.

Thank You

Explainable Healthcare AI to Save Lives at the Bedside

MSME Idea Hackathon 6.0

Theme: **Frontier Technologies (Sub-Theme 6)**

Anonymous Project Submission

Host Institute Code: MSME-HI-REDACTED

Questions & Discussion